Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
\sum_j q_a^j(|X_j|^2-|Y_j|^2)\eta_j^+\overline{\eta}_j^+ &= r_a^2\eta_a^+\overline{\eta}_a^+ && (a)\\
\sum_j q_a^j(X_j\overline{Y}_j)\eta_j^+\eta_j^+ &= 0 && (b)\\
\sum_j q_a^j(\overline{X}_jY_j)\overline{\eta}_j^+\overline{\eta}_j^+ &= 0. && (c)
\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$P=\left(\begin{array}{ccccc}
P_{(+++)} & & & \text{\Huge 0} & \\
& P_{(+-+)} & & & \\
& & P_{(+--)} & & \\
& \text{\Huge 0} & & \ddots & \\
& & & & P_{(---)}
\end{array}\right)$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left(\begin{array}{cc}\mu_1 & 0\\ 0 & \mu_2\end{array}\right)\left(\begin{array}{cc}0 & d_1\\ d_2 & 0\end{array}\right)\left(\begin{array}{cc}\bar{\mu_1}^{-1} & 0\\ 0 & \bar{\mu_2}^{-1}\end{array}\right)=\left(\begin{array}{cc}0 & 1\\ 1 & 0\end{array}\right).$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
&[q^-,\alpha_0^+]=i\,,\\
&[\alpha_n^-,\alpha_m^+]=n\,\delta_{n+m,0}\ \ ,\\
&[s_n\,,s_m]=\left(\tfrac{k}{2}-1\right)\,n\,\delta_{n+m,0}\ \ ,\\
&[L_n'\,,L_m']=(n-m)L_{n+m}'
\end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
\Sigma + \eta >& \frac{1}{M} \sum_{l=1}^{M} \Big((h^{D_1}(k_0)\phi_l, \phi_l) \|\psi_l\|_{L^2}^2 + (h^{D_2}(K-k_0)\psi_l, \psi_l) \|\phi_l\|_{L^2}^2 \Big) \\
\geq & (z_1(k_0) + z_2(K-k_0)) \, \frac{\sum\limits_{l=1}^{M} \|\phi_l\|_{L^2}^2 \|\psi_l\|_{L^2}^2}{M} = z_1(k_0) + z_2(K-k_0).
\end{aligned}
$$